

TEACHER GUIDE

Teach the companion

Swinging Sign · Stage 4 Technology

Identity and scope

This guide accompanies the 50-page student workbook for the school-developed Swinging Sign unit in Stage 4 Technology, Years 7–8. Its focus is Materials and production processes within Technology 7–8 Syllabus (2023). It follows five current modules and all 15 named learning sections. It does not claim whole-syllabus coverage or NESAs approval.

The source guide describes five weeks at five 60-minute lessons per fortnight. The teacher program organises 15 flexible learning blocks; these are not 15 promised periods. Confirm the local timetable, delivery year and practical allocation. Fifty workbook pages describe the print edition, not a course-hour entitlement.

The teacher supplies practical authority

No original working drawing, approved construction plan or job sheet was located in the frozen course and supplied source collection. Issue the current local plan before affected practical work. Dimensions, stock, materials, joints, machinery, finishing products and timing, mounting, loads and test methods remain teacher-controlled. Current demonstrations, school controls, supervision and explicit permission govern participation.

Swinging Sign is the primary pathway. The teacher may select the LED tea-light holder and scrolled pot plant holder as alternatives or additions. Tea-light use is LED-only. Each pathway needs its own confirmed support, context and use criteria.

Formative evidence and editions

Workbook questions and folio entries support learning; they do not create marks, weighting, due dates or practical sign-off. Use the current approved assessment notification for formal requirements. Print the student PDF on A4, duplex on the long edge: 50 pages use 25 sheets. Local Word edits can change pagination, so recheck page references and writing space before printing.

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Use the learning sequence

Model a reason, then collect the student's evidence.

Read, discuss and apply

Introduce the named section reading, inspect its visual and model one reasoning move. Then use the linked paper case or activity before collecting a folio response. The book supplies fictional cards and learning models so students can work on paper while the teacher presents the course. All 15 website sections also have no-stream routes; the Onshape clips are optional.

Keep CAD and production separate

Onshape is the current visible CAD teaching. Establish the school document, sharing, units, naming, version and drawing requirements before a digital session. Ask learners to explain the relationship a constraint controls and how a sketch affects a feature. A paper clinic can demonstrate understanding; it is not proof that a student completed a CAD operation or received production approval.

Adjust access while keeping the reasoning

For support, preteach a few terms, highlight the observation in a case and rehearse an answer aloud. Let students label a sketch before developing a paragraph, or use an approved scribe or speech-to-text adjustment. Sentence starters can connect feature, reason and expected effect. Keep actual, planned and unknown information distinct.

For extension, request a competing explanation, a different valid nesting arrangement or a limit on the evidence. Compare a proposed change against the same criterion. Extend through decision quality rather than invented loads, machine settings or unsupported material choices. Stop at unresolved local practical information while continuing the unaffected paper learning.

Use the answer support

References below follow each student page's keyed tasks in reading order, including drawings and table response prompts. Accept equivalent reasoning for open tasks. The 20 printed MCQs are original formative adaptations; the website retains its larger 150-question bank with feedback and section responses. This guide answers every printed keyed exercise and supplies the purpose of each folio strand.

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Locate the learning

Modules 1–3 • all page references are student pages.

Current named section	Teaching	Application / evidence
m01-s01 • Read the product and user	5–6	11–12, 45
m01-s02 • Criteria, constraints and project pathways	4, 7–8	11, 45–46
m01-s03 • Workshop readiness and sustainability	9–10	11, 48
m02-s01 • Vector and bitmap	13	17, 19, 28
m02-s02 • Onshape sketches and features	14	16–17, 20, 47
m02-s03 • Legibility, annotation and peer review	15	17–18, 20, 46–47
m03-s01 • Material properties and selection	21	25–26, 28, 47
m03-s02 • Measurement, layout and resource efficiency	22	25, 27
m03-s03 • Onshape drawings, mounting and approval	23	24–25, 28, 47–48

The current section index supplies these identities. Older URLs and asset names can retain “Inkscape”, while the visible lessons teach Onshape. Use the current title and content, not a legacy identifier, to select teaching material.

The source program links this unit to TE4-SDP-01, TE4-PDP-01, TE4-MS-01, TE4-DES-01, TE4-PPM-01 and TE4-SAF-01. These address sustainability, design/production practices, materials and systems, communication/evaluation, project processes and safe participation. Follow the program’s existing mapping; the workbook adds no whole-course claim.

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Locate checks and puzzles

Modules 4–5 · quick references

Current named section	Teaching	Application / evidence
m04-s01 · Production sequence and authorisation	29	33–34, 36, 48
m04-s02 · Edges, finishing and quality	30	32–33, 36, 49
m04-s03 · Nesting, offcuts and process evidence	31	33, 35–36, 48
m05-s01 · Surface preparation and coatings	37	40–41, 44, 49
m05-s02 · Hanging, stability and use	38	41–44, 49
m05-s03 · Evaluate, feedback and improve	39	41, 43–44, 50

Printed MCQ key · questions 1–4

Student page	Correct choices in order
11	B · D · A · C
17	A · B · C · D
25	C · D · A · B
33	B · C · D · A
41	D · A · B · C

All five supplementary puzzle purposes are represented: observation/inference/unknown on page 6; vocabulary matching on 28; broad sequencing on 34; finish states on 40; and evidence/opinion/unknown on 43. These are paper adaptations with worked reasoning in this guide, not a formal test.

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Follow the twelve folio strands

The final record brings earlier learning together.

The website's twelve evidence purposes are distributed through the module work and consolidated on pages 45–50. Retain useful earlier evidence rather than asking students to manufacture a second account. A planned case is not a completed practical record.

Current folio strand	Student evidence pages
1. Design brief and context	4, 7, 45
2. Product research	6, 12, 45
3. Criteria and constraints	7–8, 45
4. Concept options	18, 20, 46
5. Final proposal and annotations	20, 47
6. Material reasoning	26, 28, 47
7. Production plan and approval checkpoint	28, 34, 36, 48
8. Process record	35–36, 48
9. Quality evidence	32, 36, 49
10. Finish and function check	40, 42, 44, 49
11. Evaluation against criteria	43–44, 50
12. Feedback and future improvement	20, 43–44, 50

Check for a source or version, an observation, a reasoned decision and an honest status where each is relevant. Attach further permitted evidence when useful. The school decides submission format and formal requirements; completion of a workbook field is not teacher sign-off.

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Answers • pages 3–7

Model reasoning and acceptable evidence

Page 3 • Task 1

Accept a consequential local detail such as stock, dimensions, process, mounting or an approved check, plus an explanation that visual appearance or a draft model does not supply local authority. Do not require invented values.

Page 4 • Task 1

The record identifies the actual teacher-directed pathway and a plausible user/context question. For the tea-light pathway use LED only. A pot support or hanging sign requires teacher-confirmed support/use conditions; no load or mounting assumption is acceptable.

Page 5 • Task 1

Accept an identified audience, a specific communication feature and a plausible link to context. For example, visitors may benefit from a clear, prominent message. Do not require invented text, dimensions or viewing distances.

Page 6 • Task 1

In order: observation; inference; needs confirmation; observation; needs confirmation. The hidden-process claim is an inference, not an established fact; treat it as unconfirmed before practical use. Stock thickness and approved load require an authorised local source. Accept students who additionally flag the inference as needing confirmation if their reasoning preserves the distinction.

Page 6 • Task 2

Accept a precise feature, a plausible likely effect and an identified user/context. Language should show uncertainty where necessary, for example a bold outline may help an audience notice the message. Do not accept guessed manufacturing facts as observations.

Page 7 • Task 1

A strong response names pathway, user, need and context without inventing practical specifications. It identifies two relevant criteria, explains how a change could benefit one and weaken another, and links the priority to the user. Decoration versus clarity is one possible example; accept other justified tradeoffs.

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Answers · pages 8–11

Model reasoning and acceptable evidence

Page 8 • Task 1

Categories in row order: criterion; confirmed constraint; question needing confirmation; confirmed constraint. The rewritten criterion should identify a quality and evidence, such as message recognition or a context-appropriate visual character. Accept other justified criteria; no unapproved test conditions.

Page 9 • Task 1

The student must confirm the current task/plan, demonstration, local controls, conditions and permission. Prior experience alone cannot establish readiness. Accept a concise explanation linking at least two checks to the changed or unverified situation.

Page 10 • Task 1

Accept relevant questions at all five stages. Examples: sourcing — Which teacher approved option meets the need, and what evidence supports its origin? Production — Could another valid layout reduce avoidable rework or retain a useful offcut? Use — What confirmed context and durability criteria apply? Maintenance — How could the product be cared for or repaired within the approved system? End of life — Which local reuse, recycling or disposal route applies? No invented practical facts or universal disposal rule is required.

Page 10 • Task 2

Accept a relevant concern, a plausible alternative, a real tradeoff and evidence needed. Connect a readiness requirement, such as an approved option, local control or teacher-confirmed process, with the life-cycle concern to explain a revised design decision. Neither appearance nor a general sustainability preference overrides the practical boundary.

Page 11 • Question 1

B. The position of the panel is visible, while a load claim needs an authorised source and approved evidence. The other choices turn unknown practical facts into unsupported claims.

Page 11 • Question 2

D. Recognition describes a quality used to judge success. The other statements describe confirmed boundaries.

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Answers • pages 11–16

Model reasoning and acceptable evidence

Page 11 • Question 3

A. Understanding current instructions, controls and permission is part of readiness. Familiarity or another person's behaviour cannot authorise the task.

Page 11 • Question 4

C. Life-cycle thinking connects several effects. A material label, colour or recyclability alone cannot establish the whole judgement.

Page 11 • Task 5

Accept any genuine misconception from the options with a reasoned correction. For example, a photograph cannot prove an approved load, or recyclability does not account for avoidable waste and durability.

Page 12 • Task 1

Each row should contain an actual visible observation and a plausible likely effect. Both references have blank panels; do not invent existing lettering. Unknowns may include mounting, load, dimensions or materials. Accept different interpretations that are clearly qualified.

Page 12 • Task 2

Accept a transferable idea such as investigating how curved forms create character or how panel shape frames an intended message. The explanation should preserve the user/context and the need for teacher approved project information.

Page 16 • Task 1

A horizontal constraint keeps the line in the intended horizontal relationship and removes rotation. A dimension using the teacher approved value controls its length. The fixed starting point already provides its reference. No numerical length should be invented.

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Answers · pages 16–17

Model reasoning and acceptable evidence

Page 16 • Task 2

A feature depends on its source geometry. Editing that geometry can update the resulting form or affect later dependent features. The ordered feature list helps a reviewer trace this relationship and inspect where the change originated; a successful update still needs review.

Page 16 • Task 3

Example version note: Practice revision B — added purposeful horizontal and approved length controls; checked remaining movement and the dependent feature. Questions might ask whether the plane/reference expresses the intended view and whether the sketch/drawing requirements and approved dimensions are correctly represented. Accept appropriate questions about version, feature behaviour or review requirements. Do not invent an approval or export instruction.

Page 17 • Question 1

A — Bitmap pixels suit the photograph's tonal detail; vector objects support editing and scaling clean lettering. Purpose and internal structure matter more than the file label.

Page 17 • Question 2

B — Investigating the competing or duplicated controls identifies the cause. Random deletion, ignoring feedback or exporting does not establish deliberate sketch behaviour.

Page 17 • Question 3

C — It connects a feature, its purpose and expected communication effect. The other options name features without explaining a decision.

Page 17 • Question 4

D — A controlled comparison helps identify whether the change improved the intended effect. Preference alone and several simultaneous changes weaken the reasoning.

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Answers • pages 18–19

Model reasoning and acceptable evidence

Page 18 • Task 1

Both original concepts use the same chosen words and motif. A identifies the intended reading order. B changes one meaningful variable such as spacing or emphasis while keeping the others sufficiently consistent for comparison. Its annotation links a feature, reason and expected effect. Accept diverse styles; judge purposeful hierarchy and reasoning rather than taste. These are communication ideas, not approved making plans.

Page 18 • Task 2

Accept self-review or peer review. The response identifies visible features and their effect on the intended order or legibility, then proposes a focused investigation linked to a criterion. It must not present peer preference as production authority or invent an observed response.

Page 19 • Task 1

A: likely bitmap, because pixelation and photographic texture support this; suitable for a contextual photograph. B: likely vector, because editable curves/points describe paths; suitable for developing lettering. C: needs more evidence, because a container name does not reveal the image's structure; inspect before recommending an editing use. D: likely vector, because the outline points are editable and scale smoothly; suitable for a clean symbol. Smoothness alone would be insufficient, but the path evidence strengthens D.

Page 19 • Task 2

Possible trace issues include excessive points, uneven contours, unwanted background, gaps, overlaps or loss of meaningful positive/negative space. A suitable question asks who owns the source and whether the intended adaptation/reuse is permitted or teacher approved. Students should keep the photograph for context when that better serves the task; tracing is not automatically necessary or approved.

Page 19 • Task 3

Each can serve a different purpose: bitmap detail communicates real-world appearance/context, while vector objects communicate clean editable lettering or shapes. A mixed document does not make every object a vector, and each asset still needs suitability and permission checks.

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Answers · pages 20–25

Model reasoning and acceptable evidence

Page 20 • Task 1

The student identifies the proposal state and shows readable evidence with meaningful annotations. Hierarchy concerns the intended order of attention; the relationship may concern spacing, geometry or contrast. Unconfirmed dimensions, materials, mounting, document or export requirements remain questions.

Page 20 • Task 2

The response fairly records actual feedback, links it to a criterion and explains the decision. An accepted or rejected suggestion can be valid if justified. For planning-only work, describe an intended comparison rather than inventing peer feedback.

Page 20 • Task 3

Expected evidence includes a clearly identified current version, controlled/reviewable geometry where applicable, annotations, a comparison with criteria and a revision note. A relevant question concerns an unresolved teacher-controlled requirement or whether the intended effect is communicated. Readiness for review is distinct from production approval.

Page 24 • Task 1

Suitable questions ask whether revision B meets the criteria after the spacing change; what approved spacing/orientation/edge requirements apply to its layout; and what mounting relationship/support/context must be confirmed before production. Accept equivalent specific questions tied to the pack.

Page 24 • Task 2

The pack contains mismatched revision evidence and unresolved layout/mounting information, so it does not establish approval of the current proposal. The B drawing or A-to-B revision note is useful for traceability. The answer should distinguish a useful file from a complete approval decision and must not invent permission.

Page 25 • Question 1

C — A justification explains why relevant evidence supports a conditional choice and acknowledges a tradeoff. Appearance, habit or unexplained scores alone do not establish suitability.

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Answers • pages 25–26

Model reasoning and acceptable evidence

Page 25 • Question 2

D — Readings may be repeatable yet inaccurate. The reference, units and intended value still need checking.

Page 25 • Question 3

A — Connected offcut value and clear communication can distinguish layouts with equal coverage. Paper efficiency never overrides actual approved production requirements.

Page 25 • Question 4

B — Approval is tied to a defined state. The change must remain traceable and be reviewed under the teacher's process before the next practical action.

Page 26 • Task 1

Flat form: stiffness/resistance to bending favours A in the supplied fictional evidence.

Manageability/support: lower mass favours B, while suitability of either support system remains unknown.

Resource use: A's connected offcut may retain more reuse value, but actual usefulness depends on confirmed identity, form, condition and the approved system. Do not infer grades, thicknesses, loads or practical methods.

Page 26 • Task 2

Either recommendation can be defensible if it ranks the confirmed needs and uses the supplied evidence.

Example: A better supports flat form and may leave a useful offcut, but its greater mass means the support system needs review. B may be preferred for lower mass if its flexibility is acceptable. Required confirmation may concern actual stock, stiffness criterion, support suitability or offcut value. These fictional facts cannot authorise a real material choice.

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Answers · pages 27–28

Model reasoning and acceptable evidence

Page 27 • Task 1

Each layout contains six non-overlapping 2×5 -cell rectangles within the 10×10 grid. One complete example fills columns 1–6 across all ten rows, splitting each two-column strip into an upper and lower rectangle; columns 7–10 remain one connected 4×10 region. A contrasting example fills columns 1–2, 5–6 and 9–10, again splitting each strip into two rectangles; the two remaining 2×10 strips are separate. Accept other valid arrangements. Outline the actual largest connected blank region. These are cell positions in a paper model, not project measurements.

Page 27 • Task 2

Each rectangle covers $2 \times 5 = 10$ cells. Six cover 60 of 100 cells, so coverage is 60% and unused area is 40% in both layouts. Equal percentages do not establish equal usefulness: shape and connectedness affect possible future use. In the model examples, one 40-cell region differs from two separate 20-cell regions. Actual offcut identity, condition, safe management and reuse require the teacher's system.

Page 27 • Task 3

The recommendation compares whether required shapes are clear, how useful the remaining region may be and a limitation of the paper model. Actual stock, orientation, spacing, edge allowances and process requirements need teacher confirmation. Either valid layout is acceptable when the reasoning is specific and does not claim production approval.

Page 28 • Task 1

Row order: constraint; vector; criterion; purpose; tradeoff. Accept a meaningful connection, such as criteria translate purpose into checkable expectations; a tradeoff balances benefits within constraints; or vector communication can serve the sign's purpose but still needs criteria-based review.

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Answers • pages 28–32

Model reasoning and acceptable evidence

Page 28 • Task 2

The response identifies a current version/sequence label and a material reason linked to a confirmed need, property and evidence. It states the actual review status — planned, awaiting review, revision requested or explicitly approved within a stated scope — and names a relevant unresolved question. If local material information is absent, a conditional argument is appropriate. No invented approval, support capacity or layout rule is acceptable.

Page 31 • Task 1

Useful output remains 60 cells (60%). The spoiled shape consumes an extra 10 cells, leaving 30 rather than 40 unused cells; the largest connected remainder falls from 40 to 20. Check the outlines and layout before committing paper, then compare correct shapes, spoiled area and connected remainder under the same rules. Accept another justified change. These model figures provide no project dimensions, allowances or permission.

Page 32 • Task 1

Example: The upper feature appears tilted compared with the approved orientation example. This may affect the required orientation/legibility. Refer the difference to the teacher to confirm the check and next action. The response should not name an unverified cause or attempt an unapproved correction.

Page 32 • Task 2

Example: The confirmed measurement of the checked feature differs from the reviewed drawing's value. The size criterion may not be met; the teacher needs to verify the reference, measurement and permitted response. No invented numerical difference or tolerance is required.

Page 32 • Task 3

Example: A mark is visible on one area of the outer face. It differs from the approved even-appearance criterion. The teacher should confirm whether the variation is acceptable or requires an authorised response. The note is criteria-linked rather than a personal preference.

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Answers · pages 33–34

Model reasoning and acceptable evidence

Page 33 • Question 1

B — Dependencies explain why a stage must precede another. They support relevant checks without prescribing a universal machine sequence.

Page 33 • Question 2

C — The uncertain condition must be referred for the teacher-controlled inspection and next action. A photograph, size impression or student guess cannot establish safe handling.

Page 33 • Question 3

D — Offcut value and safe management require several confirmed facts and the teacher's system. Appearance alone does not establish a handling or reuse decision.

Page 33 • Question 4

A — These records connect plan, authorisation, outcome and response. The other sets provide little traceable evidence of the decisions that controlled production.

Page 34 • Task 1

The broad order is C, D, E, A, B. Accept loops back for revision. Suitable hold points include proposal review before material commitment, a produced-feature quality check before finishing, and confirmation of readiness/method before testing. Students prepare/record; the teacher checks/authorises; only the locally authorised person performs practical action. The diagram must not invent machine steps or imply the student has authority to operate.

Page 34 • Task 2

The affected work pauses before relying on the changed feature. The current design/version, change reason and affected criteria, layout, material/mounting relationship or later-stage dependencies return for teacher review. Approval of the earlier state cannot be silently transferred to the revised state. The exact local next action is teacher-controlled.

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Answers • pages 35–36

Model reasoning and acceptable evidence

Page 35 • Task 1

Select A, C, F, H and K, normally in that order. Sample captions: A — Dated layout V2 records the proposed arrangement. C — The review record identifies V2 and the particular next stage authorised. F — The actual arrangement follows authorised production using V2. H — The quality note compares the result with an approved reference. K — The offcut record shows referral through the teacher's system. Do not invent a result absent from the card, broaden the approval or select the unapproved personal information in I.

Page 35 • Task 2

Accept a specific change such as clear version/date labels, captions captured at hold points, or a check for a complete plan-to-result sequence. Explain how it improves traceability, privacy or relevance and how the next folio would be reviewed to see whether the problem decreased. More photographs without a purpose is not sufficient.

Page 36 • Task 1

The evidence is the student's own permitted record or a clearly labelled planned view. It identifies the stage and relevant feature/reference. It must not claim a hidden check, completed practical work or teacher observation that did not occur.

Page 36 • Task 2

Expected response links the stage to the actual plan/version and an approved criterion or reference, then reports a specific observation. Differences or missing evidence may be recorded honestly. Planning-only work must remain in future/conditional language.

Page 36 • Task 3

The student states the actual proceed/revise/stop decision, decision-maker and limited scope of the next action. If no decision is available, record awaiting teacher review. The resource note concerns avoidable material use, planned versus actual layout, rework or an offcut referral through the teacher's system. It must not invent permission to handle, reuse or dispose of material.

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Answers · pages 40–41

Model reasoning and acceptable evidence

Page 40 • Task 1

Bare surface; primer stage; topcoat; teacher to confirm. Layer purposes can be discussed conceptually, but actual products and application methods require local confirmation.

Page 40 • Task 2

Supported: a primer has a preparatory role; the topcoat provides the intended outer surface. These purpose labels do not establish actual suitability. Unsupported: compatibility with the substrate/layers/environment is assumed; feeling dry is treated as proof of readiness for testing. Do not infer product identity or curing time.

Page 40 • Task 3

Questions may ask which substrate/system is approved, what evidence confirms compatibility for the outdoor context, and what product/teacher information establishes readiness for the next stage. A suitable inspection statement identifies an observable criterion and an honest status, such as checking intended coverage against the approved appearance reference when authorised. Do not claim an inspection or test has already occurred.

Page 41 • Question 1

D — Different film-development states have different meanings. Actual readiness must come from current product information and teacher direction, not a universal time or appearance assumption.

Page 41 • Question 2

A — A load path describes the route of forces through the connected system. It is conceptual reasoning, not a specification for a particular mounting or load.

Page 41 • Question 3

B — A conclusion must stay within the scope of the evidence. A classroom check does not authorise new locations, loads or public installation.

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Answers • pages 41–42

Model reasoning and acceptable evidence

Page 41 • Question 4

C — It names a purposeful change and a relevant comparison. A complete proposal would explain the expected benefit and the evidence to collect. Vague effort, preference and concealment do not provide a testable improvement.

Page 42 • Task 1

The first drawing places equal cards at equal distances on opposite sides of the central suspension. Its stated level result is consistent with balanced effects in this simplified model. The second keeps positions fixed and identifies greater mass on one side; that side tends downward. Load-path arrows link each card through the connection/bar to the support: they describe transfer through the system, not measured force vectors. Do not infer a final angle, support capacity, permitted load or actual installation.

Page 42 • Task 2

Greater mass on one side changes mass distribution and the relative effect about the support, so the heavier side tends downward in the simplified case. A different change to connection position, distance from support, an opening or an attachment could also alter orientation. Accept another source-consistent relationship; no numerical force calculation is required.

Page 42 • Task 3

Do not transfer support capacity, connector suitability, dimensions, loads, acceptable movement, weather performance or installation/test methods. The teacher must confirm the actual product-support system, location, loads, criteria, approved method and authorisation before hanging/loading/use. Tea-light use remains LED-only; any pot-holder pathway needs its own confirmed pot/load and stability requirements.

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Answers · pages 43–44

Model reasoning and acceptable evidence

Page 43 • Task 1

The intended categories are evidence; opinion; unknown/confirm; evidence; opinion. The edge and level records support only their stated checks and context, not every hidden condition, future load or public installation. “Every viewer will prefer this colour” is an unsupported preference generalisation. Also accept unknown/confirm for that statement if the learner explains that no evidence about every viewer is supplied; do not penalise a defensible distinction between taste and an unverified universal claim.

Page 43 • Task 2

A and C are criterion-linked/useful evidence comments: A concerns legibility; C concerns traceability. B is preference-only unless linked to a relevant criterion through further evidence. D needs clarification. A and C are the strongest because each identifies a specific issue and direction.

Page 43 • Task 3

Accept a specific spacing comparison under the same approved viewing conditions, or preserving labelled versions with a check that a reviewer can trace the decision. The pitch should identify the issue, describe one feasible change, predict a relevant benefit and state evidence that would show whether it helped. Do not invent practical dimensions, results or assessment marks.

Page 44 • Task 1

The record is the student’s own permitted evidence and identifies the stage and feature. A planning-only or missing-evidence state is acceptable when clear. A photograph cannot prove hidden strength, unrestricted stability or long-term finish performance; annotations should stay within what was actually checked.

Page 44 • Task 2

The judgement cites actual observations/approved test results, identifies a supported strength and a limitation or not-yet-evidenced criterion. Any classification depends on the criterion, effect and decision history. A planned record must not report a fictional success. Formal assessment and permissions remain teacher-controlled.

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Answers • pages 44–46

Model reasoning and acceptable evidence

Page 44 • Task 3

The student fairly records relevant feedback, explains an accept/modify/reject response, and proposes a specific product/process/evidence improvement. It predicts a benefit and describes an appropriate future check within teacher authority. The transferable lesson is a principle, such as testing legibility early or keeping versions traceable, not copied dimensions, material specifications or operating procedures.

Page 45 • Task 1

The actual confirmed pathway and a coherent brief are required. A sign brief identifies its communication purpose; alternate pathways identify the relevant holding/support purpose. No unapproved dimensions, methods or loads.

Page 45 • Task 2

Require three specific criteria linked to the brief, each with a suitable evidence proposal. For a sign, possible pairs are message recognition with the approved viewing record; suitable orientation with the teacher approved system check; and intended surface appearance with comparison to the approved reference. Accept different relevant criteria or an alternative pathway. The teacher must approve actual methods and conditions; these examples do not prescribe a practical test.

Page 45 • Task 3

Distinguish confirmed local requirements from pending questions and identify where confirmation came from. A photograph or unsupported assumption is not an authorised source.

Page 46 • Task 1

A complete concept should respond to the chosen pathway, audience and purpose. Labels connect visible decisions with intentions and confirmed constraints. Exact construction facts remain teacher-controlled.

Page 46 • Task 2

The second option should make a meaningful change to arrangement, form, hierarchy or another permitted feature. Accept different valid designs; do not require copied reference geometry.

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Answers · pages 46–48

Model reasoning and acceptable evidence

Page 46 • Task 3

A sound comparison uses the same criterion for both options, records a specific feedback point and explains a proposed revision. Preference alone is insufficient; feedback can be accepted or declined with a reason.

Page 47 • Task 1

The proposal is identifiable and relevant to the confirmed pathway. Native annotations should explain feature, reason and intended effect. A pasted image needs meaningful annotation and an identifiable version. Practical specifications remain teacher-controlled.

Page 47 • Task 2

Accept the actual identifiable document/drawing/version used, with a specific review point. A fictional filename, assumed approval or anonymous screenshot does not establish the record.

Page 47 • Task 3

Look for an approved option, a correct relevant property/use relationship and a competing consideration such as mass, durability, appearance, resource use or maintenance. A material cannot be identified definitively by appearance alone.

Page 47 • Task 4

The response names relevant missing plan, version, stock, dimensions, process, support or permission details. The teacher's actual review determines readiness; the workbook record is not approval.

Page 48 • Task 1

Use the actual teacher approved dependencies. Four broad record rows could group proposal/review; authorised production and monitoring; approved finishing/readiness; and approved testing/evaluation. Useful evidence might include the current version and review decision, a quality observation, the confirmed finish/readiness record and the actual criterion-linked result. These are possible groupings, not a compulsory local sequence. Additional stages may be attached. No machine instructions or settings should be invented.

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Answers • pages 48–49

Model reasoning and acceptable evidence

Page 48 • Task 2

Record an actual review outcome and conditions, or an honest pending status. This record must not invent teacher approval or replace the school's authorisation process.

Page 48 • Task 3

A credible process entry identifies real work, current version/stage, an observation, decision and reason. A photo reference supports rather than replaces explanation. If practical work has not occurred, record not yet completed.

Page 48 • Task 4

Accept a real decision about layout, usable offcuts, rework, materials or another resource, supported by an observation. Avoid declaring waste savings without an actual comparison.

Page 49 • Task 1

Each completed row needs an actual criterion or reference, approved method/conditions and observed evidence. Surface appearance alone does not prove curing, hidden quality or load capacity. No practical test is authorised by this table.

Page 49 • Task 2

A useful detail is authentic and locatable. It can show a surface, edge, clearance or other approved feature; annotation must distinguish visible evidence from unknown hidden performance.

Page 49 • Task 3

The explanation links evidence to a criterion and a justified action. Any practical correction must have the required authorisation. A pending check or unresolved question is a valid honest outcome.

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Answers · page 50

Model reasoning and acceptable evidence

Page 50 • Task 1

A balanced evaluation names two relevant criteria and connects observations to a strength and limitation. Missing evidence limits the claim; it must not be replaced with an imagined successful test.

Page 50 • Task 2

Feedback is specific and relevant. The improvement responds to the observed limitation, explains an expected effect and proposes an appropriate check for teacher approval. Accept a justified decision not to adopt a suggestion.

Page 50 • Task 3

Accept a specific reflection about observation, comparison, control of a model, planning, quality or evaluation, linked to something the student actually did or reconsidered.

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Source notes and decisions

Snapshot retained for this workbook · 12 September 2026

Current course and version

Matching course: <https://stevenowell.github.io/Swinging-Sign/> . The frozen repository revision is bc4699f4dd13f69c41c1b2415f36628351f4c686. The current section index, fifteen section extracts, website answer key, activities, puzzles and twelve-card folio establish the learning scope. Exact source files and byte digests are retained with the build.

Source conflicts resolved

The older teacher Drive guide supports the broad five-week design-to-evaluation sequence. Its Inkscape/CNC proposals do not override the current visible Onshape lessons. Its proposed assessment split is not imported as a formal scheme. The current assessment page records the absence of an approved notification; the original-plan finding records that no approved construction drawing or job sheet was supplied.

The teacher program manifest supplies the Stage 4 Technology identity, 15 flexible blocks, six mapped outcomes and local confirmation requirements. The authority-decision record explains the current CAD choice, practical limits, primary and alternate pathways, privacy, photograph permissions and offline equivalence. The course release-state record does not grant permission to publish or install a student product.

Official framework

<https://curriculum.nsw.edu.au/learning-areas/tas/technology-7-8-2023/overview/course>

The retained official outcome and Materials and production processes content pages corroborate the framework. This workbook neither creates new outcome claims nor substitutes for approved local sources and protocols where additional syllabus content is taught.

TEACHER GUIDE

Use the illustrations honestly

Visual provenance and limits

What the artwork represents

The edition combines newly generated cover and pathway artwork, existing same-course generated teaching illustrations, the privacy-safe blank-plaque reference and an original vector-bitmap concept diagram. Generated images illustrate ideas; they are not photographs of completed student work or evidence that a teacher observed, approved or tested a product. No exact image-engine version is asserted here.

The source asset inventory and source ledger record provenance and reuse limits. The two original teacher supplied holder photographs have permission limited to the source course release and are excluded from direct workbook reproduction. The original personalised sign photograph is not used to identify a student; the illustrative blank reference supports discussion of form and purpose.

Teach the visible relationship

Read each figure with its native caption. Material-like appearance does not establish material identity or grade. A generated CAD screen is not an authentic Onshape interface or approved drawing. A surface image cannot certify edge safety, coating compatibility or curing. A hanging-product image cannot establish connection capacity, weather suitability or permission to install it.

The paper nesting grids and balance case state their own learning assumptions. Grid cells are not project dimensions. A balanced model is not a safe-load result for the real sign. Keep hypothetical or planned records separate from authentic evidence, and use only permitted photographs when students document work.

Maintain the evidence trail

The structured student content is the common basis for the PDF, editable Word edition and this answer guide. Preserve task references when making local changes and recheck the corresponding answers, page references and practical boundaries. Mirror Edition 2.0, Footstool and Desk Tidy inform editorial presentation only; their project facts are not construction sources for Swinging Sign.